
UCL COMP0078: Supervised Learning

Problem Sheet

Online learning I

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If you find any mistakes/typos, please contact sl-support@cs.ucl.ac.uk.

Overview of Problems

- **Problem 1:** Develop a mistake-bound algorithm for learning a sequence consistent with a k -literal conjunction.
- **Problem 2:** Explore the relationship between the perceptron mistake bound for linearly separable data and the hard margin of support vector machines.
- **Problem 3:** Demonstrate how to establish a regret bound for the weighted majority algorithm.
- **Problem 4:** Provide an example showcasing the lower bound of mistake rates, illustrating how to obtain lower bounds for mistake rates.
- **Problem 5:** Illustrate how to analyse randomization predicted weighted majority algorithm.

Problem 1

Suppose $\mathcal{X} = \{\text{True}, \text{False}\}^n$. Give a polynomial time algorithm A with a mistake bound $\mathcal{M}(A) \leq O(n^2)$ for any example sequence which is consistent with a k -literal conjunction. Your answer should contain an argument that $\mathcal{M}(A) \leq O(n^2)$.

Solution 1

We take the perceptron algorithm for simplicity. We have the following mistake bound:

$$\mathcal{M} \leq R^2 \|\mathbf{u}\|^2$$

where $R^2 = \max_t \|\mathbf{x}_t\|^2$, and this holds for any $\mathbf{u}$ for which $\langle \mathbf{u}, \mathbf{x}_t \rangle y_t \geq 1$. To apply this to the conjunction problem, we make use of de Morgan's Law, which states that $x \wedge y = \overline{(\bar{x} \vee \bar{y})}$, and represent the conjunction as a disjunction of the negated variables. We take S to be the set of indices of the variables in the conjunction, and define

$$u_i = \begin{cases} -2 & \text{if } i \in S \\ 1 & \text{if } i = n+1 \\ 0 & \text{otherwise} \end{cases}.$$

Instead of considering $\mathbf{x} \in \{0, 1\}^n$, we take the feature map $\phi(\mathbf{x}) = (\bar{\mathbf{x}}, 1)$, where $\bar{x}_i = \begin{cases} 1 & \text{if } x_i = 0 \\ 0 & \text{if } x_i = 1 \end{cases}$.

Then we have that $R^2 \leq n+1$, and $\|\mathbf{u}\|^2 = 4k+1$, giving

$$\mathcal{M}(A) \leq (n+1)(4k+1).$$

Problem 2

State the perceptron convergence theorem [Novikoff] explaining the relation with the hard margin support vector machine solution.

Solution 2

The number of mistakes made by the perceptron algorithm is:

$$\mathcal{M} \leq \frac{R^2}{\gamma^2},$$

if there exists a vector $\mathbf{v}$ that satisfy $\langle \mathbf{v}, \mathbf{x}_t \rangle y_t \geq \gamma$ and $\|\mathbf{v}\| = 1$. The γ term is analogous to the SVM margin term.

Problem 3

Define the c -regret of learning algorithm as

$$c\text{-regret}(m) = L_A(S) - c \min_{i \in [n]} L_i(S)$$

thus the usual regret is the 1-regret.

1. Argue for the weighted majority set-up argue that without randomised prediction it is impossible for all training sequences to obtain sublinear c -regret for $c < 2$.
2. Show how to select β to obtain sublinear 2-regret.

Solution 3.1

We consider a training sequence with two experts E_1 and E_2 that give constant predictions, so that for all $t \in [T]$

$$\begin{aligned} x_{t,1} &= 0 \\ x_{t,2} &= 1 \end{aligned}$$

We also consider an adversarial environment that forces mistakes for each prediction. Then, $\min_{i \in \{1,2\}} \mathcal{M}_i \leq \frac{T}{2}$, but $\mathcal{M}_A = T$. Hence we have that $\mathcal{M}_A \geq 2 \min_{i \in \{1,2\}} \mathcal{M}_i$

Solution 3.2

Starting with

$$\mathcal{M} \leq \frac{\ln(\frac{1}{\beta})}{\ln(\frac{2}{1+\beta})} \mathcal{M}_i + \frac{1}{\ln(\frac{2}{1+\beta})} \ln(n)$$

Defining $\epsilon := 1 - \beta$, and assuming that $\epsilon \in [0, 1/2]$ so that we can use the inequality $-\log(1 - \epsilon) \leq \epsilon + \epsilon^2$ for $\epsilon \in [0, 1/2]$, we obtain

$$\ln\left(\frac{1}{\beta}\right) = -\ln(1 - \epsilon) \leq \epsilon + \epsilon^2$$

and using the inequality $-\log(1 + x) \geq -x$

$$\ln\left(\frac{2}{1+\beta}\right) = -\ln\left(\frac{1+\beta}{2}\right) = -\ln\left(1 + \frac{\epsilon}{2}\right) \geq \frac{\epsilon}{2}$$

thus giving

$$\begin{aligned}\mathcal{M} &\leq 2(1 + \epsilon)\mathcal{M}_i + \frac{2}{\epsilon} \ln(n) \\ &\leq 2\mathcal{M}_i + 2\epsilon T + \frac{2}{\epsilon} \ln(n)\end{aligned}$$

From the AM-GM inequality, we know that for $\frac{x+y}{2} \geq \sqrt{xy}$ and is only equal when $x = y$. Therefore,

$$2\epsilon T = \frac{2}{\epsilon} \ln(n) \implies \epsilon = \sqrt{\frac{\log(n)}{T}}$$

so tuning $\epsilon = \sqrt{\frac{\log(n)}{T}}$, and assuming that $\sqrt{\frac{\log(n)}{T}} \leq \frac{1}{2}$ we then

$$\mathcal{M} \leq 2\mathcal{M}_i + 4\sqrt{\log(n)T}$$

In the case that $\sqrt{\frac{\log(n)}{T}} > \frac{1}{2}$, we can write the generic bound

$$\begin{aligned}\mathcal{M} &\leq T \\ &\leq 2\mathcal{M}_i + T \\ &< 2\mathcal{M}_i + 4\log(n)\end{aligned}$$

Combining the 2 cases, we then have

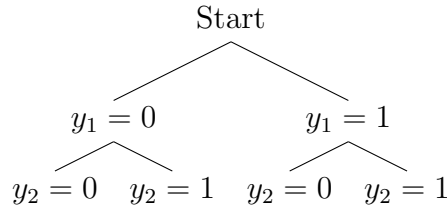
$$\mathcal{M} \leq 2\mathcal{M}_i + 4\sqrt{\log(n)T} + 4\log(n).$$

Problem 4

Consider binary prediction with expert advice, with a perfect expert. Prove that any algorithm makes at least $\Omega(\min(T, \log_2 n))$ mistakes in the worst case.

Solution 4

Let's assume that $\log_2 n$ is an integer. Consider the case where we have experts that give each of the following predictions in the tree:



Defining N to be the tree-depth $- 1$, we have 2^N experts, and N mistakes can be forced when $N \geq T$, and T mistakes if $N < T$. Taking $n = 2^N$, we then have $\min(T, \log_2(n))$ forced mistakes.

Problem 5

Recall that by tuning the weighted majority we achieved a bound

$$\mathcal{M} \leq 2.63 \min_i \mathcal{M}_i + 2.63 \ln n.$$

Now by using randomisation in the prediction, design an algorithm with a bound that has the property

$$\frac{\mathcal{M}}{T} \leq \min_{i \in [n]} \frac{\mathcal{M}_i}{T}$$

as $T \rightarrow \infty$, for the weighted majority setting (i.e., the mean the prediction error of the algorithm is bounded by the mean prediction error of the “best” expert). Recalling that m is the number of examples (and the “tuning” of the algorithm may depend on m). For contrast compare this to problem 3.1 above.

Solution 5

Initialise weights of all experts to be 1. We consider the following protocol at each timestep t :

1. Receive expert predictions $\mathbf{x}_t \in \{0, 1\}^n$
2. Predict $\hat{y}_t = \begin{cases} 0 & \text{with probability } \frac{\sum_{i: x_{t,i}=0} w_{t,i}}{W_t} \\ 1 & \text{otherwise} \end{cases}$.
3. Receive $y_t \in \{0, 1\}$
4. Weights of incorrect experts multiplied by $\beta \in [0, 1)$.

Define A_t to be the proportion of experts that give the wrong prediction at time t . Note that the $\mathbb{E}[\mathcal{M}] = \sum_{t=1}^T A_t$. We have

$$\begin{aligned}
 W_{T+1} &= W_T((1 - A_T) + A_T\beta) \\
 &= W_T(1 - (1 - \beta)A_T) \\
 &= W_1 \prod_{t=1}^T (1 - (1 - \beta)A_t) \\
 &\leq n \prod_{t=1}^T \exp(-(1 - \beta)A_t) \\
 &= n \exp\left(-\sum_{t=1}^T (1 - \beta)A_t\right) \\
 &= n \exp(-(1 - \beta)\mathbb{E}[\mathcal{M}])
 \end{aligned}$$

where the inequality comes from $\exp(-x) \geq 1 - x$. Since we also have that $W_{T+1} = \sum_{i=1}^n \beta^{M_i} \geq \beta^{M_i}$ for any $i \in [n]$, we have

$$\beta^{M_i} \leq n \exp(-(1 - \beta)\mathbb{E}[\mathcal{M}]).$$

Taking logs, we obtain

$$M_i \log(\beta) \leq \log(n) - (1 - \beta)\mathbb{E}[\mathcal{M}].$$

Rearranging,

$$\mathbb{E}[\mathcal{M}] \leq \frac{-\log \beta}{1 - \beta} M_i + \frac{1}{1 - \beta} \log(n)$$

Defining $\epsilon := 1 - \beta$ and assuming that $\epsilon \in [0, 1/2]$ so that we can use the inequality $-\log(1 - \epsilon) \leq \epsilon + \epsilon^2$ for $\epsilon \in [0, 1/2]$, the following follows

$$\begin{aligned}
 \mathbb{E}[\mathcal{M}] &\leq (1 + \epsilon)M_i + \frac{1}{\epsilon} \log(n) \\
 &\leq M_i + \epsilon T + \frac{1}{\epsilon} \log(n)
 \end{aligned}$$

From the AM-GM inequality, we know that for $\frac{x+y}{2} \geq \sqrt{xy}$ and is only equal when $x = y$. Therefore,

$$2\epsilon T = \frac{2}{\epsilon} \ln(n) \implies \epsilon = \sqrt{\frac{\log(n)}{T}}$$

so tuning $\epsilon = \sqrt{\frac{\log(n)}{T}}$ which is in $[0, 1/2]$ for $T \geq \log(n)$ gives

$$\mathbb{E}[\mathcal{M}] - M_i \leq 2\sqrt{\log(n)T}.$$

$$\frac{\mathbb{E}[\mathcal{M}]}{T} - \frac{M_i}{T} \leq 2\sqrt{\frac{\log(n)}{T}}.$$

The upper bound vanishes as $T \rightarrow \infty$.